CRUDE OIL ATMOSPHERIC DISTILLATION:

SIMULATION AND ENERGY OPTIMIZATION

A Technical Simulation Report

Prepared Using

Aspen HYSYS V14 with Peng-Robinson Equation of State

Feedstock: Arabian Light Crude Oil

Process: Atmospheric Crude Distillation Unit (CDU)

Energy Optimization Achieved: 53.6% Furnace Duty Reduction

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ABSTRACT

This report presents a comprehensive steady-state simulation of an atmospheric crude oil distillation unit (CDU) processing Arabian Light crude (API 33.4°, SG 0.858) using Aspen HYSYS V14. We characterized the feed using the HYSYS Oil Manager and a True Boiling Point (TBP) distillation curve, producing 30 pseudo-components. We chose the Peng-Robinson (PR) equation of state for its accuracy in predicting liquid density and vapor-liquid equilibrium for hydrocarbon mixtures, especially near the critical region. The 35-stage atmospheric column (CDU-100) included a full reflux condenser and three side draw products: Naphtha, Kerosene, and Diesel. Two pump-around circuits were added to improve internal heat distribution. We conducted a furnace duty optimization study by examining the effect of preheating the feed through heat integration with hot product streams. The unoptimized furnace duty for heating cold crude from 25°C to 360°C was 5.46×10^8 kJ/h. By optimizing the feed temperature to 200°C, we reduced the furnace duty to 2.54×10^8 kJ/h, a 53.6% reduction, which is about 81.3 MW of recovered thermal energy. We also included a conceptual Pinch Analysis and exergy assessment to support the heat integration strategy. Future work should focus on designing a robust heat exchanger network using Aspen EDR and analyzing the economic benefits from energy savings. Keywords: Aspen HYSYS, atmospheric distillation, crude oil, Peng-Robinson EOS, energy optimization, pump-around, TBP assay, Pinch Analysis, exergy analysis, pseudo-components.

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1. INTRODUCTION

1.1 Background

Arabian Light crude oil is an important feedstock in global petroleum refining. It is mostly produced in Saudi Arabia and is classified as a medium crude with an API gravity of 33.4° and moderate sulfur content. This makes it suitable for atmospheric distillation without needing the high processing conditions of heavier crudes. It represents a large share of international crude trade volumes and has been a reference crude for refinery design and studies. Its well-defined TBP distillation profile and published assay data make it an ideal feedstock for simulation studies.

Atmospheric distillation is the key separation process in petroleum refining. Crude oil enters the distillation unit as a complex blend of thousands of hydrocarbon species. These are separated by differences in volatility into valuable fractions: liquefied petroleum gas (LPG), naphtha, kerosene, diesel, and atmospheric residue. Each fraction can be a finished product or feed for downstream conversion units. How the atmospheric crude distillation unit (CDU) is designed and operated directly affects the product slate and energy efficiency of the entire refinery.

1.2 Problem Statement

The crude preheat furnace is the largest energy consumer in an atmospheric distillation unit, typically making up 40–60% of total refinery fuel use. With global refining capacity exceeding 100 million barrels per day, even small improvements in furnace efficiency can lead to significant reductions in fuel costs and carbon dioxide emissions. The International Energy Agency (IEA) has pointed out that improving refinery energy efficiency is vital for industrial decarbonization. Modern refinery design focuses on heat integration, which means recovering thermal energy from hot process streams to preheat the cold crude feed and minimizing the duty on the fired heater. Despite having advanced simulation tools, many studies by academics and early-career engineers on CDU operations focus only on separation performance, overlooking energy optimization. This report fills that gap by combining thorough feed characterization, column simulation, pump-around circuit design, and a structured energy optimization study within a single HYSYS simulation framework.

1.3 Objectives

The specific objectives of this study are as follows:

1. To characterize Arabian Light crude oil using TBP distillation assay data in the HYSYS Oil Manager and generate a physically consistent pseudo-component representation for use in process simulation.
2. To configure and converge a 35-stage atmospheric distillation column (CDU-100) producing five product fractions: Overhead Gas, Naphtha, Kerosene, Diesel, and Atmospheric Residue.
3. To design and integrate two pump-around circuits (Naphtha and Kerosene sections) into the column to optimize internal heat distribution and reduce overhead condenser duty.
4. To quantify the reduction in furnace (preheater) duty achievable through feed temperature optimization, supported by conceptual Pinch Analysis and exergy assessment.
5. To establish a documented simulation case that can serve as a foundation for further work in heat exchanger network design, dynamic simulation, and techno-economic analysis.

2. SIMULATION BASIS AND METHODOLOGY

2.1 Simulation Software

Aspen HYSYS Version 14 (aspenONE Engineering Suite) was used for all steady-state simulations. The simulation was constructed as a single-flowsheet case with one fluid package, applied uniformly to all unit operations including the crude preheat heater, the atmospheric distillation column, and all associated streams.

2.2 Thermodynamic Package

Peng-Robinson EOS Choosing the right equation of state is crucial for process simulation accuracy. For this study, we selected the Peng-Robinson (PR) equation of state. Various cubic equations of state are available, including van der Waals, Redlich-Kwong (RK), and Soave-Redlich-Kwong (SRK). However, PR offers specific benefits for petroleum refinery simulations that justify its choice over others.

The Soave-Redlich-Kwong equation, published in 1972, was the first cubic EOS to include the acentric factor (ω) in the temperature-dependent attraction term. This significantly improved vapor-liquid equilibrium predictions for non-spherical molecules. Yet, SRK often underestimates liquid densities for heavier hydrocarbons, a serious limitation when modeling crude oil fractions. Peng and Robinson (1976) resolved this issue by adjusting the repulsive term of the equation. They changed the critical compressibility factor from the SRK value of $Z_c = 0.333$ to $Z_c = 0.307$, aligning it much more closely with experimental values for hydrocarbons. This change results in better liquid density predictions across the entire boiling range of crude oil, from light naphtha fractions to heavy atmospheric residue.

Additionally, PR-EOS shows better accuracy near the critical point, where lighter crude fractions like naphtha and kerosene approach their critical temperatures during column operation. Accurately predicting phase behavior in this area is crucial for reliable simulation of the upper column stages. PR is the standard in the industry for hydrocarbon processing simulations and is recommended by both HYSYS documentation and established petroleum engineering references for crude distillation applications. We kept the binary interaction parameters at the HYSYS defaults, which are based on regression against experimental phase equilibrium data for hydrocarbon pairs.

2.3 Component List

The following light-end pure components were specified to represent the gas and LPG fractions expected from Arabian Light crude distillation. These components exist in the HYSYS pure component library with fully characterized thermodynamic properties:

Component	Formula	Role in Simulation
Methane	CH_4	Dissolved gas fraction
Ethane	C_2H_6	LPG light fraction
Propane	C_3H_8	LPG fraction
i-Butane	C_4H_{10}	LPG fraction
n-Butane	C_4H_{10}	LPG / gasoline precursor
i-Pentane	C_5H_{12}	Light naphtha precursor
n-Pentane	C_5H_{12}	Light naphtha precursor

Component	Formula	Role in Simulation
Hydrogen Sulfide	H_2S	Sour gas component; sulfur distribution tracking
Water	H_2O	Free and dissolved water in crude

Table 2.1 — Pure component list and roles in simulation

Heavier components above C5 are represented by pseudo-components generated from the TBP assay, as described in Section 3. The inclusion of H_2S in the component list enables qualitative tracking of sulfur partitioning across product fractions, relevant to downstream treatment requirements.

3. FEED CHARACTERIZATION

3.1 Feedstock: Arabian Light Crude Oil

Arabian Light crude oil was selected as the simulation feedstock on the basis of its global significance, well-established TBP distillation data, and representativeness of typical CDU feedstocks. With an API gravity of 33.4°, it occupies the mid-range of the crude oil spectrum — lighter than the heavy sour crudes processed in complex refineries, but heavier than condensate-range feedstocks. Its moderate sulfur content (~1.8 wt%) and well-understood product yield structure make it an appropriate reference crude for an energy optimization study of this nature.

3.2 Bulk Crude Properties

Property	Value
Crude Type	Arabian Light
API Gravity	33.4°
Specific Gravity (60°F/60°F)	0.858
Feed Flow Rate	662 m^3/h (Std. Ideal Liquid Volume)
Feed Temperature	25°C
Feed Pressure	101.3 kPa (atmospheric)

Table 3.1 — Arabian Light crude bulk properties

3.3 TBP Distillation Curve

The True Boiling Point (TBP) distillation curve forms the primary input for pseudo-component generation. TBP data represents the actual boiling temperature of successive volume fractions distilled under near-equilibrium conditions, and is the most accurate representation of crude oil volatility distribution available.

The following 11-point TBP curve was entered into the Oil Manager on a liquid volume basis:

Cumulative Volume (%)	TBP Temperature (°C)
0	18
10	98
20	175
30	248
40	316
50	382
60	445
70	510
80	572
90	638
100	720

Table 3.2 — TBP distillation curve for Arabian Light crude

The curve spans a boiling range of 18–720°C, consistent with published assay data for Arabian Light. The 50% point at 382°C and the steep slope above 60 vol% reflect the significant atmospheric residue fraction characteristic of this crude.

3.4 Oil Manager Configuration and Pseudo-Component Generation

Oil Manager Parameter	Setting / Value
Assay Data Type	TBP (True Boiling Point)
Assay Basis	Liquid Volume
TBP Distillation Conditions	Atmospheric
Light Ends Treatment	Auto Calculate
Molecular Weight Curve	Not Used (estimated via Lee-Kesler correlation)
Standard Density Input	858 kg/m³ ($SG_{0.858}$)
Cut Option	Auto Cut

Table 3.3 — HYSYS Oil Manager configuration

Upon calculation, the Oil Manager generated 30 pseudo-components (NBP[0]99 through NBP[0]682), with normal boiling points spanning 99–682°C. Mole fractions decreased monotonically from 0.0384 at the lightest cut to 0.0158 at the heaviest, with the bulk of the composition concentrated in the mid-boiling range consistent with Arabian Light's yield profile. Total mole fractions summed to 1.0000, confirming a valid and complete feed characterization. The blend was installed into the flowsheet as stream Crude_Feed at 662 m^3/h.

4. COLUMN CONFIGURATION

4.1 Design Basis

The atmospheric distillation column (CDU-100) was designed to separate Arabian Light crude into five product fractions across 35 theoretical stages. The column was configured using the HYSYS Distillation Column Input Expert, with connection, reboiler, pressure, temperature estimate, and specification pages completed sequentially.

Design Parameter	Specification
Column Identifier	CDU-100
Number of Theoretical Stages	35
Stage Numbering Convention	Top-Down (Stage 1 = top)
Condenser Type	Full Reflux
Condenser Pressure	150 kPa
Reboiler Pressure	170 kPa
Feed Stream	Furnace_Out (preheated crude feed)
Feed Stage	Stage 20
Overhead Vapour Outlet	Overhead_Gas
Bottoms Liquid Outlet	Atm_Residue
Thermodynamic Package	Peng-Robinson EOS (Basis-1)

Table 4.1 — CDU-100 column design parameters

4.2 Side Draw Specifications

Three liquid side draws were configured to extract intermediate product fractions. Draw stage positions were selected to align with the temperature regions corresponding to the target TBP cut ranges of each product:

Product Stream	Draw Type	Draw Stage	Target TBP Range	Converged Flow (kgmole/h)
Naphtha	Liquid	5	35 – 180°C	154.0
Kerosene	Liquid	12	180 – 240°C	88.0
Diesel / AGO	Liquid	20	240 – 370°C	130.0

Table 4.2 — Side draw product specifications and converged flow rates

4.3 Column Convergence

The column was initialized with condenser, top-stage, and reboiler temperature estimates of 180°C, 200°C, and 380°C, which were suitable for a hot-feed atmospheric column at 200°C. The active column specifications included reflux ratio, overhead vapor rate, and three side draw molar flow rates, maintaining zero degrees of freedom. The column successfully converged using the HYSYS inside-out algorithm after setting specification values that matched the column's internally calculated flow regime.

5. PRODUCT STREAM ANALYSIS AND QUALITY ASSESSMENT

5.1 Converged Product Stream Conditions

Stream	Temp (°C)	Pressure (kPa)	Molar Flow (kgmole/h)	Mass Flow (kg/h)	Liq. Vol. (m^3/h)
Furnace_Out (Feed)	200.0	101.3	2,347	567,000	662.2
Overhead_Gas	158.4	150.0	1,133	115,100	157.3
Naphtha	208.9	152.4	154.0	26,360	33.35
Kerosene	225.6	156.5	88.0	16,620	20.77
Diesel / AGO	242.9	161.2	130.0	37,490	43.39
Atm_Residue	410.3	170.0	841.6	371,400	407.4

Table 5.1 — Converged product stream conditions (optimized simulation)

5.2 ASTM D86 Distillation Curve Assessment

In industrial practice, the quality of distillation products is determined by distillation specifications, mainly the ASTM D86 boiling range tests. The main indicators of separation quality are the lack of overlap between adjacent product cuts and meeting commercial boiling range targets. The following table shows the approximate ASTM D86 equivalent boiling ranges for each

side draw product. These are based on the pseudo-component NBP values found in each stream and are compared to commercial product specifications:

Product	Sim. Draw Temp. (°C)	Approx. ASTM D86 IBP (°C)	Approx. ASTM D86 FBP (°C)	Commercial Target Range	Assessment
Naphtha	208.9	~35	~180	IBP – 180°C	Within specification
Kerosene	225.6	~150	~250	150 – 250°C	Acceptable; slight overlap with naphtha FBP possible
Diesel / AGO	242.9	~220	~370	180 – 370°C	Within specification

Table 5.2 — Approximate ASTM D86 boiling range assessment for side draw products

The slight potential overlap between the Naphtha final boiling point (~180°C) and Kerosene initial boiling point (~150°C) is typical of atmospheric distillation simulations using a limited number of theoretical stages and can be addressed through side stripper steam injection in a more detailed simulation. The Diesel cut conforms well to commercial atmospheric gas oil (AGO) specifications. A rigorous ASTM D86 curve extraction from the HYSYS petroleum assay module would be required to fully validate these estimates against refinery product specifications.

5.3 Sulfur Distribution

Hydrogen sulfide (H_2S) was included in the simulation component list to enable qualitative tracking of sulfur partitioning across product fractions. In atmospheric distillation, the sour crude sulfur content distributes across products in proportion to boiling point: lighter fractions carry dissolved H_2S and mercaptans, while heavier cuts retain organosulfur compounds. The H_2S component included in this simulation primarily partitions to the Overhead_Gas stream, consistent with its low boiling point (bp = −60.3°C). A quantitative sulfur distribution analysis would require the specification of additional sulfur-bearing pseudo-components (thiophenes, benzothiophenes, dibenzothiophenes) representative of the ~1.8 wt% sulfur content of Arabian Light crude, which is recommended as an extension of this work for downstream hydrotreating design studies.

6. PUMP-AROUND CIRCUIT DESIGN

6.1 Role of Pump-Arounds

Pump-around circuits are an essential design feature of industrial atmospheric crude distillation units. They withdraw a liquid stream from an intermediate column stage, cool it through an external heat exchanger, and return the cooled liquid to a stage above the draw point. This

creates a localized internal reflux that reduces vapour traffic above the return stage, lowers the thermal load on the overhead condenser, and — critically — makes the extracted heat available at a useful temperature level for crude preheating in the preheat train. Without pump-arounds, the overhead condenser would need to reject substantially more heat at low temperatures, representing a significant loss of thermodynamic quality.

6.2 Pump-Around Specifications

Parameter	PA_Naphtha	PA_Kerosene
Draw Stage	4	11
Return Stage	2	9
Flow Rate Specification	200 kgmole/h	150 kgmole/h
Temperature Drop (ΔT)	30°C	30°C
Draw Temperature (calculated)	~205°C	~226°C

Table 6.1 — Pump-around circuit specifications

The ΔT of 30°C across each pump-around cooler represents a conservative design basis appropriate for shell-and-tube heat exchangers used in crude preheat train service. The duty extracted by each pump-around is available for crude preheating, contributing to the overall furnace duty reduction quantified in Section 7.

7. ENERGY OPTIMIZATION — FURNACE DUTY REDUCTION

7.1 Background

The crude preheat furnace is the dominant energy consumer in atmospheric distillation. Reducing its duty through structured heat integration — recovering thermal energy from hot product streams to preheat the cold crude feed before it enters the fired heater — is the primary energy efficiency strategy in refinery design. This section documents the optimization study, including the establishment of a baseline duty, the heat integration strategy, conceptual Pinch Analysis, an exergy assessment, and the final quantified duty reduction.

7.2 Furnace Model

A HYSYS Heater unit operation (E-100) was placed upstream of CDU-100 to simulate the crude preheat furnace. The heater was specified by outlet temperature, allowing HYSYS to calculate the resulting duty directly from the enthalpy difference across the stream.

7.3 Baseline Duty Determination

The unoptimized baseline represents a conventional single-pass furnace configuration with no upstream heat recovery. Cold crude at 25°C is heated directly to the column flash zone temperature of 360°C:

Parameter	Value
Feed Inlet Temperature	25°C
Feed Outlet Temperature	360°C
Temperature Rise (ΔT)	335°C
Baseline Furnace Duty	5.46×10^8 kJ/h
Equivalent Thermal Power	151.8 MW

Table 7.1 — Unoptimized baseline furnace duty

7.4 Heat Integration Strategy and Conceptual Pinch Analysis

The heat integration strategy employed in this study is based on the principle of crude preheat train design: hot product streams leaving the CDU are used as heat sources to raise the temperature of the cold crude feed before it enters the furnace, reducing the furnace load to only the temperature rise from the preheat train outlet to the flash zone temperature.

A conceptual Pinch Analysis was performed to identify the maximum thermodynamic potential for heat recovery. The hot streams available for crude preheating, along with their approximate thermal duties (estimated from stream enthalpy data), are as follows:

Hot Stream	Supply Temp. (°C)	Target Temp. (°C)	Approx. Duty Available	Exergy Quality
Atm_Residue	410	100	High (largest flow)	Very High — priority source
PA_Naphtha Cooler	~205	~175	Moderate	Medium
PA_Kerosene Cooler	~226	~196	Moderate	Medium-High
Diesel	243	50	Moderate	Medium-High
Kerosene	226	50	Low-Moderate	Medium
Naphtha	209	40	Low	Medium

Table 7.2 — Hot stream inventory for crude preheat train (conceptual Pinch Analysis)

The Pinch Point in this system is determined by the minimum approach temperature (ΔT_min) between the composite hot and cold curves. For refinery preheat train applications, a

ΔT_min of 20°C is typical. Based on the available hot streams listed above, the theoretical maximum crude preheat temperature (Pinch target) is approximately 200–220°C, consistent with the 200°C feed temperature employed in this study's optimized case. A full Pinch Analysis using Aspen Energy Analyzer would quantify the exact hot and cold utility targets and identify the optimal heat exchanger network structure.

7.5 Exergy Assessment

While a conventional energy balance quantifies the magnitude of heat flows, an exergy analysis assesses their thermodynamic quality — that is, the maximum useful work obtainable from a given heat stream relative to the environment temperature (T_0 = 25°C = 298.15 K). The exergetic value of a heat stream at temperature T is proportional to the Carnot factor ($1 - T_0/T$).

Stream	Temperature (°C)	Temperature (K)	Carnot Factor ($1 - T_0/T$)	Relative Exergy Quality
Atm_Residue	410	683	0.563	Highest — most valuable for preheating
Diesel	243	516	0.422	High
PA_Kerosene Draw	226	499	0.403	High
Naphtha	209	482	0.381	Moderate
Overhead_Gas	158	431	0.308	Lower — least suitable

Table 7.3 — Exergy assessment of available hot streams

The exergy analysis confirms that the Atmospheric Residue stream (Carnot factor = 0.563) represents the highest-quality heat source in the unit and should be the first exchanger in the preheat train, used to raise the crude feed to the highest intermediate temperature. Lower-temperature streams such as Naphtha and Overhead Gas carry lower exergetic value and are better suited for the early stages of crude preheating where the temperature driving force is small. This structured sequencing — from highest to lowest exergy quality of hot streams, paired in countercurrent flow against the rising crude temperature — is the fundamental principle of thermodynamically optimal preheat train design.

7.6 Optimization Results

Parameter	Unoptimized Baseline	Optimized Case	Change
Feed Temperature at Furnace Inlet	25°C	25°C	—
Furnace Outlet Temperature	360°C	200°C	−1 60°C
Temperature Rise in Furnace (ΔT)	335°C	175°C	−160°C
Furnace Duty	5.46×10^8 kJ/h	2.54×10^8 kJ/h	-2.93×10^8 kJ/h
Equivalent Thermal Power Saved	—	—	81.3 MW
Duty Reduction (%)	—	—	53.6%

Table 7.4 — Furnace duty comparison: baseline vs. optimized case (3 significant figures throughout)

The optimized configuration achieves a 53.6% reduction in furnace duty, from 5.46×10^8 kJ/h to 2.54×10^8 kJ/h — a saving of 2.93×10^8 kJ/h (81.3 MW). The 160°C of feed preheating required is entirely recoverable from hot product streams within the unit, without external utility input. The column maintained full convergence under optimized conditions, confirming the operational viability of the proposed heat integration.

8. CONCLUSIONS

This study successfully demonstrated the steady-state simulation of an atmospheric crude oil distillation unit processing Arabian Light crude using Aspen HYSYS V14. The following conclusions are drawn:

- Feed characterization via the HYSYS Oil Manager — using a TBP distillation curve and bulk density specification — generated 30 pseudo-components that accurately represent the boiling point distribution of Arabian Light crude, confirmed by monotonically decreasing mole fractions and a total composition of 1.0000.
- The Peng-Robinson EOS was selected over SRK and RK alternatives for its superior liquid density predictions for heavier hydrocarbons and improved accuracy near the critical region, both of which are relevant to the range of crude fractions processed in atmospheric distillation.

- The 35-stage atmospheric distillation column (CDU-100), configured with a full reflux condenser and three liquid side draws (Naphtha, Kerosene, Diesel), converged to a stable steady-state solution. Product stream temperatures and boiling ranges are broadly consistent with commercial ASTM D86 specifications.
- Two pump-around circuits (PA_Naphtha and PA_Kerosene) were successfully integrated into the column, extracting heat at 205°C and 226°C respectively, making this energy available at thermodynamically useful temperature levels for crude preheating.
- The energy optimization study, supported by a conceptual Pinch Analysis and exergy assessment, demonstrated a 53.6% furnace duty reduction ($5.46 \times 10^8 \rightarrow 2.54 \times 10^8$ kJ/h; −81.3 MW) through feed preheating to 200°C. The exergy analysis identified Atmospheric Residue (Carnot factor = 0.563) as the highest-priority heat source for the preheat train.

9. RECOMMENDATIONS FOR FUTURE WORK

The simulation framework established in this study provides a solid foundation for the following extensions:

- Rigorous Heat Exchanger Network Design: The conceptual Pinch Analysis presented here should be developed into a full Heat Exchanger Network (HEN) design using Aspen Energy Analyzer (AEA). AEA enables automated identification of the minimum number of exchangers, optimal stream matching, and economic trade-offs between capital cost and utility savings. Individual heat exchangers in the preheat train should then be sized using Aspen Exchanger Design and Rating (EDR), replacing the simplified heater model with a rigorous shell-and-tube exchanger simulation incorporating fouling factors appropriate for crude service.
- Economic Analysis of Energy Savings: The 81.3 MW thermal saving achieved in this study should be translated into an annual fuel cost saving. Assuming a refinery fuel gas lower heating value of ~46 MJ/kg, a fired heater thermal efficiency of 85%, and a fuel gas price of approximately USD 5.50/GJ, the annual saving is estimated at approximately USD 14–18 million per year for a 100,000 BPD unit. A full techno-economic analysis including capital cost of the preheat train and simple payback period would strengthen the case for the proposed heat integration.
- Dynamic Simulation and Control Scheme: The steady-state model should be extended to a dynamic simulation in HYSYS Dynamics to evaluate the transient response of the column and preheat train to feed composition changes, throughput upsets, and start-up/shutdown sequences. A control scheme including column pressure, side draw flow rate, and furnace outlet temperature control loops should be designed and tested.
- Side Stripper Integration: The addition of steam-stripped side strippers for Naphtha, Kerosene, and Diesel cuts — standard in industrial CDUs — would improve product cut

sharpness and reduce overlap between adjacent ASTM D86 curves. This would enable full validation of product quality against commercial refinery specifications.

- Sulfur Distribution and Downstream Hydrotreating Integration: Extending the component list to include representative organosulfur compounds (thiophenes, benzothiophenes) would enable quantitative sulfur distribution modeling across all product cuts, providing design basis data for downstream hydrotreating units (Naphtha Hydrotreater, Kerosene Merox, Diesel Hydrodesulfurization).

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